

AI and the Future of Software Engineering

Topic 8: Real Industry Case Studies

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Introduction

Artificial intelligence has had a very large and rapid impact in recent years all over the world. But one of the most transformed industries is undoubtedly that of software engineering. The reformation of this industry through artificial intelligence has brought about the automation of processes in a shorter time and has made them less costly. This has brought greater income to the respective companies, but has also reduced the number of employees. AI today is being used not only to facilitate processes, but also to detect problems, predict them and analyze the market and its demands.

This study focuses on concrete cases of companies that have implemented artificial intelligence in their work. The analysis focuses on how these companies use AI, how they have changed their work processes, the benefits they have received and the challenges they have encountered

1. Microsoft

How Microsoft Uses AI

Microsoft is one of the companies that is at the forefront of implementing AI across all of its products, platforms, and research initiatives. In its consumer and enterprise products, Microsoft integrates AI to improve user experiences, exemplified by features in Windows 11, GitHub Copilot, and Office 365. Microsoft has also participated in the creation of some of the protocols related to artificial intelligence that are used by many other companies.

Workflow Changes and Advantages

Before AI, developers at every company and at Microsoft spent a lot of time writing repetitive code. The adoption of AI tools like GitHub Copilot has led to a significant increase in the speed and productivity of software development, with up to 30% of the company's code now generated by AI. These tools have also helped to find errors and improve developer performance as they focus not on syntax but on logical solutions. Microsoft is also contributing to the development of a more open and collaborative AI ecosystem that encourages interoperability and scalability across platforms.

- Microsoft is pioneering AI agents and multi-agent orchestration using open standards like A2A.
- Copilot+ PCs bring AI to the edge, enabling real-time, on-device intelligent experiences.
- Microsoft Research is infusing AI into scientific discovery, data visualization, and responsible AI evaluation.
- Microsoft's AI Skills Fest democratizes AI literacy for learners across skill levels and geographies.
- Open collaboration is key that Microsoft supports protocols like MCP and A2A to drive interoperability.

Limitations and Challenges

Despite the positive aspects, Microsoft is also facing problems related to the artificial intelligence included in its products. Ensuring that artificial intelligence systems are consistent with human values, safe and explainable remains a major concern, especially as these systems become more autonomous and influential in decision-making. The question for this giant in the industry remains how independent should artificial intelligence be and how dependent should we be on it?

2. Google

How Google Uses AI

Google is one of the leading companies in artificial intelligence research and implementation. The company integrates AI into many of its products and services such as Google Search, Google Cloud, Android, YouTube, and Google Workspace. Google also develops advanced AI models through DeepMind and Gemini AI systems.

In software engineering, Google uses AI to assist developers in code generation, bug detection, software testing, and infrastructure management. AI systems help engineers manage very large-scale systems with millions of lines of code and improve the efficiency of development processes.

Workflow Changes and Advantages

Before AI integration, Google developers spent significant time manually reviewing code, identifying bugs, and monitoring infrastructure systems. With the implementation of AI-

powered tools, many of these repetitive tasks became partially automated. AI systems now help developers by suggesting code improvements, detecting possible software bugs, automating testing procedures, monitoring cloud infrastructure and improving software maintenance processes.

Google engineers can now focus more on system architecture, innovation and solving complex technical problems instead of repetitive operational work.

- Google uses Gemini AI models to support software development and productivity tools.
- DeepMind contributes advanced AI research for automation and intelligent systems.
- AI-powered infrastructure monitoring helps detect unusual system behavior faster.
- Google Cloud integrates AI solutions for developers and enterprise customers.
- AI improves scalability and efficiency across Google's global platforms.
- Google promotes responsible AI development through safety and ethical research initiatives.

Limitations and Challenges

Despite its advancements, Google also faces several challenges regarding artificial intelligence integration. AI systems may sometimes produce inaccurate predictions or false results, which can affect software quality and reliability. Privacy and data security remain important concerns because Google handles enormous amounts of user information.

Another major challenge is balancing automation with human supervision. Although AI can automate many engineering tasks, human developers are still necessary to verify results, ensure ethical use, and make critical technical decisions. The rapid growth of AI also raises concerns about energy consumption, computational costs, and the future impact of automation on software engineering jobs.

3. Shopify

How Shopify Uses AI

Shopify is increasingly integrating artificial intelligence into both software engineering and business operations. The company uses AI tools to help developers improve coding efficiency, automate repetitive tasks, and optimize workflows. Shopify also applies AI in customer support systems, product recommendations, and e-commerce analytics.

The company encourages employees and developers to adapt to AI-powered workflows and modern development tools. AI assistants help developers generate code faster and improve productivity during software development.

Workflow Changes and Advantages

Before the use of AI tools, Shopify developers spent more time writing repetitive code and manually handling operational tasks. With AI integration, many development processes became faster and more efficient. Developers now use AI for code generation, debugging assistance, workflow automation, customer support systems and improving development speed.

AI tools allow software engineers to focus more on product innovation, user experience, and strategic improvements instead of repetitive implementation work.

- Shopify uses AI-powered tools to improve software development productivity.
- AI assists in automating customer support and e-commerce operations.
- Developers use AI coding assistants to reduce repetitive programming tasks.
- AI-driven recommendation systems improve customer experiences on Shopify platforms.
- Shopify promotes adaptation to AI technologies across engineering teams.

Limitations and Challenges

Despite the advantages, Shopify also faces several challenges regarding AI integration. AI-generated code may contain errors or security vulnerabilities if developers do not carefully review outputs. Another concern is the growing dependence on AI systems, which could reduce developers' independent problem-solving skills over time.

Maintaining software quality, security, and ethical AI usage remains an important responsibility for Shopify's engineering teams. Human supervision is still necessary to verify AI-generated results and ensure reliable software development.

4. Duolingo

How Duolingo Uses AI

Duolingo is one of the companies that actively uses artificial intelligence to improve both software engineering and user learning experiences. The company integrates AI into language learning systems, lesson generation, chatbot conversations, and personalized education tools.

Duolingo recently expanded its use of generative AI to create educational content more efficiently and support interactive language learning features. AI systems help developers and content teams improve scalability and deliver personalized experiences to millions of users worldwide.

Workflow Changes and Advantages

Before AI integration, Duolingo teams spent significant time manually creating exercises and educational content. With AI-powered systems, much of the repetitive content generation process became automated. Developers and content creators now use AI for generating language exercises, creating chatbot interactions, personalizing learning experiences, automating explanations and improving application performance.

AI-assisted workflows allow Duolingo to release updates and educational content faster while supporting a growing global user base.

- Duolingo uses generative AI for educational content creation.
- AI-powered chatbots improve language learning interactions.
- Personalized AI systems adapt lessons to individual users.
- AI helps Duolingo scale its platform more efficiently.
- Software engineers use AI tools to accelerate development workflows.

Limitations and Challenges

Despite the benefits, Duolingo also faces challenges with AI integration. AI-generated educational content may sometimes contain inaccuracies or language mistakes. Ensuring educational quality and maintaining user trust remain major concerns.

Another challenge involves balancing automation with human expertise. While AI can accelerate content creation, human educators and engineers are still necessary to review and validate the generated material. Ethical concerns also exist regarding the replacement of human work with automated systems.

Common Trends Across Companies

Several important patterns appear across all case studies.

AI Assists Rather Than Replaces

In all companies, AI primarily supports developers instead of replacing them completely. Human engineers still make important decisions, review code, design systems, and solve complex problems.

Productivity Improvement

All companies reported productivity improvements after integrating AI tools. Developers can complete repetitive tasks faster and focus more on creative or strategic work.

Human Supervision Remains Necessary

AI systems are not fully reliable. Incorrect code generation, hallucinations, security risks, and software quality concerns require continuous human supervision.

Changing Software Engineering Skills

Modern software engineers increasingly need:

- critical thinking
- system design understanding
- AI tool usage skills
- debugging abilities
- communication skills

Knowing how to work with AI tools is becoming an important professional skill.

Challenges of AI Integration

Although AI provides many advantages, companies also face significant challenges.

Security Risks

AI-generated code can introduce vulnerabilities if developers do not carefully review outputs.

Overdependence on AI

Developers may become too dependent on AI assistance and reduce their own problem-solving abilities.

Ethical Concerns

AI raises ethical questions regarding:

- job displacement
- copyright
- ownership of generated code
- fairness in automation

Reliability Problems

AI systems may produce incorrect or misleading outputs. Engineers must still validate results carefully.

Conclusion

Real industry case studies show that AI is already changing software engineering workflows in major technology companies. Microsoft, Google, Shopify, and Duolingo all use AI to improve productivity, automate repetitive tasks, and accelerate development processes.

However, AI does not eliminate the need for software engineers. Instead, the role of engineers is evolving. Human skills such as critical thinking, system design, verification, creativity, and communication remain essential.

The future of software engineering will likely involve collaboration between humans and AI systems rather than full automation. Engineers who learn how to use AI responsibly while maintaining strong technical fundamentals will remain highly valuable in the industry.

